

```
>_MONGOSH
> use day2
< switched to db day2
> db.patients.insertMany([
  { patientName: "Arun", doctor: "Dr. Ravi", department: "Cardiology", age: 45, treatmentCost: 20000, daysAdmitted: 5 },
  { patientName: "Meena", doctor: "Dr. Kumar", department: "Neurology", age: 50, treatmentCost: 30000, daysAdmitted: 7 },
  { patientName: "Raj", doctor: "Dr. Ravi", department: "Cardiology", age: 60, treatmentCost: 15000, daysAdmitted: 3 },
  { patientName: "Divya", doctor: "Dr. Mehta", department: "Orthopedics", age: 35, treatmentCost: 10000, daysAdmitted: 2 },
  { patientName: "Kiran", doctor: "Dr. Kumar", department: "Neurology", age: 40, treatmentCost: 25000, daysAdmitted: 6 },
  { patientName: "Sneha", doctor: "Dr. Mehta", department: "Orthopedics", age: 30, treatmentCost: 8000, daysAdmitted: 2 },
  { patientName: "Vijay", doctor: "Dr. Ravi", department: "Cardiology", age: 55, treatmentCost: 22000, daysAdmitted: 4 },
  { patientName: "Priya", doctor: "Dr. Kumar", department: "Neurology", age: 48, treatmentCost: 27000, daysAdmitted: 5 },
  { patientName: "Anil", doctor: "Dr. Mehta", department: "Orthopedics", age: 52, treatmentCost: 12000, daysAdmitted: 3 },
  { patientName: "Riya", doctor: "Dr. Ravi", department: "Cardiology", age: 29, treatmentCost: 18000, daysAdmitted: 4 }
])
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```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a1183658031788622976719'),
    '1': ObjectId('6a118365803178862297671a'),
    '2': ObjectId('6a118365803178862297671b'),
    '3': ObjectId('6a118365803178862297671c'),
    '4': ObjectId('6a118365803178862297671d'),
    '5': ObjectId('6a118365803178862297671e'),
    '6': ObjectId('6a118365803178862297671f'),
    '7': ObjectId('6a1183658031788622976720'),
    '8': ObjectId('6a1183658031788622976721'),
    '9': ObjectId('6a1183658031788622976722')
  }
}
> //Patients name and department whose cost is greater than 20000
> db.patients.aggregate([
  {$match:{treatmentCost:{$gt:20000}}},
  {$project:{patientName:1,department:1,_id:0}}
])
< {
  patientName: 'Meena',
  department: 'Neurology'
```

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```
> //Patients sorted by treatment in Descending order
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> db.patients.aggregate([
  {$sort:{treatmentCost:-1}}
])
```

```
< {
  _id: ObjectId('6a118365803178862297671a'),
  patientName: 'Meena',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 50,
  treatmentCost: 30000,
  daysAdmitted: 7
}
{
  _id: ObjectId('6a1183658031788622976720'),
  patientName: 'Priya',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 48,
  treatmentCost: 27000,
  daysAdmitted: 5
}
{
```

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```
> //First 5 patient records
> db.patients.aggregate([
  {$limit:5}
])
< {
  _id: ObjectId('6a1183658031788622976719'),
  patientName: 'Arun',
  doctor: 'Dr. Ravi',
  department: 'Cardiology',
  age: 45,
  treatmentCost: 20000,
  daysAdmitted: 5
}
{
  _id: ObjectId('6a118365803178862297671a'),
  patientName: 'Meena',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 50,
  treatmentCost: 30000,
  daysAdmitted: 7
}
{
```

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>_MONGOSH
>
> //Skip first 3 patients
> db.patients.aggregate([
  {$skip:3}
])
< {
  _id: ObjectId('6a118365803178862297671c'),
  patientName: 'Divya',
  doctor: 'Dr. Mehta',
  department: 'Orthopedics',
  age: 35,
  treatmentCost: 10000,
  daysAdmitted: 2
}
{
  _id: ObjectId('6a118365803178862297671d'),
  patientName: 'Kiran',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 40,
  treatmentCost: 25000,
  daysAdmitted: 6
}
{
```

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```
> //Patients from Cardiology department
> db.patients.aggregate([
  {$match:{department:"Cardiology"}},
  {$project:{patientName:1,_id:0}}
])
< {
  patientName: 'Arun'
}
{
  patientName: 'Raj'
}
{
  patientName: 'Vijay'
}
{
  patientName: 'Riya'
}
> //Total treatment cost per department
> db.patients.aggregate([
  {$group:{_id:"$department",totalCost:{$sum:"$treatmentCost"}}}
])
< {
  id: 'Neurology',
```

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```
> //Avg treatment cost per doctor and result in ascending order
> db.patients.aggregate([
  {$group: {_id: "$doctor", avgCost: {$avg: "$treatmentCost"}}},
  {$sort: {avgCost: 1}}
])
< {
  _id: 'Dr. Mehta',
  avgCost: 10000
}
{
  _id: 'Dr. Ravi',
  avgCost: 18750
}
{
  _id: 'Dr. Kumar',
  avgCost: 27333.333333333332
}
> //Department where total treatment cost is greater than 50000
> db.patients.aggregate([
  {$group: {_id: "$department", totalCost: {$sum: "$treatmentCost"}}},
  {$match: {totalCost: {$gt: 50000}}}
])
< {
```

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>_MONGOSH
>
> //Number of patients in each department in descending order
> db.patients.aggregate([
  {$group: {_id: "$department", count: {$sum: 1}}},
  {$sort: {count: -1}}
])
< {
  _id: 'Cardiology',
  count: 4
}
{
  _id: 'Orthopedics',
  count: 3
}
{
  _id: 'Neurology',
  count: 3
}
> Top 3 doctors with highest total treatment cost
❌ > SyntaxError: Missing semicolon. (1:3)

> 1 | Top 3 doctors with highest total treatment cost
    | ^
> //Top 3 doctors with highest total treatment cost
```

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```
> //Top 3 doctors with highest total treatment cost
> db.patients.aggregate([
  {$group: {_id: "$doctor", totalCost: {$sum: "$treatmentCost"}}},
  {$sort: {totalCost: -1}},
  {$limit: 3}
])
< {
  _id: 'Dr. Kumar',
  totalCost: 82000
}
{
  _id: 'Dr. Ravi',
  totalCost: 75000
}
{
  _id: 'Dr. Mehta',
  totalCost: 30000
}
> //Patient age > 40, group department, count no: of patients.
> db.patients.aggregate([
  {$match: {age: {$gt: 40}}},
  {$group: {_id: "$department", count: {$sum: 1}}}
])
```

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```
> //Avg treatment cost per department
> db.patients.aggregate([
  {$group: {_id: "$department", avgCost: {$avg: "$treatmentCost"}}},
  {$project: {department: "$_id", avgCost: 1, _id: 0}},
  {$sort: {avgCost: -1}}
])
< {
  avgCost: 27333.333333333332,
  department: 'Neurology'
}
{
  avgCost: 18750,
  department: 'Cardiology'
}
{
  avgCost: 10000,
  department: 'Orthopedics'
}
> //Departments having more than 2 patients admitted
> db.patients.aggregate([
  {$group: {_id: "$department", count: {$sum: 1}}},
  {$match: {count: {$gt: 2}}}
])
```

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> //Top 3 departments based on highest total treatment cost after skipping one

```
> db.patients.aggregate([
  {$group: {_id: "$department", totalCost: {$sum: "$treatmentCost"}}},
  {$sort: {totalCost: -1}},
  {$skip: 1},
  {$limit: 3}
])
```

< {

 _id: 'Cardiology',

 totalCost: 75000

}

{

 _id: 'Orthopedics',

 totalCost: 30000

}

> //Top 2 doctors treating patients with cost more than 10000

```
> db.patients.aggregate([
  {$match: {treatmentCost: {$gt: 10000}}},
  {$group: {_id: "$doctor", totalCost: {$sum: "$treatmentCost"}}},
  {$sort: {totalCost: -1}},
  {$limit: 2}
])
```

< {

```
> _MONGOSH
{
  _id: 'Cardiology',
  totalCost: 75000
}
{
  _id: 'Orthopedics',
  totalCost: 30000
}
> //Top 2 doctors treating patients with cost more than 10000
> db.patients.aggregate([
  {$match:{treatmentCost:{$gt:10000}}},
  {$group: {_id:"$doctor",totalCost:{$sum:"$treatmentCost"}}},
  {$sort:{totalCost:-1}},
  {$limit:2}
])
< {
  _id: 'Dr. Kumar',
  totalCost: 82000
}
{
  _id: 'Dr. Ravi',
  totalCost: 75000
}
day2> |
```